

MID TERM EXAMINATION**B. TECH PROGRAMMES (UNDER THE AEGIS OF USICT)****THIRD SEMESTER, NOVEMBER 2024****Paper Code: CIC-205 • Subject: Discrete Mathematics****Time: 1.5 Hours****Max. Marks: 30**

Note: Attempt Question No. 1 which is compulsory and any two more questions from remaining. (Prime notation ' denotes Boolean complementation).

Q. No.	Question	Max. Marks	CO(s)
1. (a)	State the Principle of Inclusion and Exclusion. Justify with a suitable example.	2	CO1
(b)	Define the Well Ordering Principle.	2	CO1
(c)	Find the converse, inverse, and contrapositive of the conditional statement: "If $3y - 2 = 10$, then $x = 1$."	2	CO1
(d)	Define a Lattice.	2	CO2
(e)	Express the following statement in symbolic logic using quantifiers: "Every student in class has studied Calculus."	2	CO1
2. (a)	If R denotes a relation on the set of all ordered pairs of positive integers defined by: (a, b) R (c, d) if and only if $ad = bc$ Show that R is an equivalence relation.	5	CO1
(b)	Find the Principal Conjunctive Normal Form (PCNF) without constructing truth table: $[P \rightarrow (Q \wedge R)] \rightarrow [\neg P \rightarrow (\neg Q \wedge \neg R)]$	5	CO1
3. (a)	Minimize the given Boolean expression using Karnaugh Map (K-Map): $f(A, B, C, D) = ABC'D' + ABC'D + ABCD + ABCD' + AB'CD'$ (where C' denotes the complement of variable C).	5	CO2
(b)	Consider the set $A = \{4, 5, 6, 7\}$. Let R be the relation $\leq$ (less than or equal to) on A. Determine: i. Relation R as a set of ordered pairs ii. Directed Graph (Digraph) of R iii. Adjacency Matrix of R iv. Hasse Diagram of (A, R) v. Determine whether (A, R) forms a Lattice or not.	5	CO2
4. (a)	What are closures of relations? Explain Reflexive, Symmetric, and Transitive closures with illustrative examples.	5	CO1
(b)	What is a Bijective function? Show whether $f(x) = x^3$ is a bijective function over $\mathbf{R} \rightarrow \mathbf{R}$ (set of real numbers).	5	CO2